PV101 Day 7 Student Worksheet

Load assessment and basic system sizing | Guided practice

Day_07_PV101_Load_Assessment_and_System_Sizing_2026-07-15

Name: _____ Date: _____

Operating rules

- Use only the values provided.
- Show formulas, numbers, and units.
- Circle assumptions.
- Check reasonableness before asking for signoff.
- This is guided practice—not a product design or proposal.

Provided training assumptions

Item	Training value
System voltage	12 V
Usable battery fraction / maximum depth of discharge	80% for this exercise
Autonomy for this exercise	1 day
Peak sun hours for this exercise	4 h/day
Loss multiplier for this exercise	1.25
Inverter headroom for this exercise	25% above concurrent AC load

Part A: RV training loads

Load	Power	Hours/day	Calculation	Wh/day
3 interior LED lights	8 W each	4 h		
Water pump	60 W	0.5 h		
Vent fan	36 W	3 h		

Phone/USB charging	24 W	2 h
Laptop charger (AC)	65 W	2 h

Total daily energy: _____ Wh/day

Part B: Storage

Required stored Wh = daily Wh x autonomy = _____

_____

Battery nameplate Wh = required stored Wh / usable fraction = _____

_____

Battery Ah at 12 V = battery nameplate Wh / 12 V = _____

_____

Reasonableness sentence: _____

_____

Part C: Array and inverter

Raw array watts = daily Wh / peak sun hours = _____ W

_____

Adjusted array watts = raw array watts x 1.25 = _____ W

_____

Concurrent AC load used for inverter sizing: _____ W

_____

Minimum inverter rating with 25% headroom: _____ W

_____

What load or assumption most affects the result? _____

_____

Part D: One-line diagram

Draw: PV -> controller -> battery -> inverter/DC loads. Show return paths, protection, disconnects, labels, and meter points.

_____

_____

_____

_____

Verification and signoff

Check	Peer	Instructor
Loads and hours copied correctly		
Every calculation shows units		
Assumptions match provided values		
Answers are reasonable and rounded clearly		
Diagram includes both directions/return path		
Protection, disconnects, labels, and test points shown		